Pretty Pelvis: A Virtual Pet Application That Breaks Sedentary Time by Promoting Gestural Interaction

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Abstract
Prolonged sitting can be detrimental to our postures which can cause spinal and pelvic problems. It may also lead to a shortened lifespan due to its adverse impact on cardio metabolic health. One way of preventing and mitigating these risks is to break sedentary time. We find it difficult, however, to actively monitor ourselves and commit to making these breaks in the middle of studying or working. The work presented in this paper proposes a motivational smartphone application that connects to a sensor-equipped seat device to track sitting behavior. The application's distinguishing feature is its interactive virtual pet, which acts as a motivational factor to promote healthier sitting habits. The application analyzes sensor readings from the seat device and prompts users with their virtual pet to take physical action. Our prototype is the outcome of a thorough investigation of user needs.

Author Keywords
Sedentary behavior; interactive chair interface; quantified-self; virtual pet; gestural interaction

ACM Classification Keywords
H.5.2. Information interfaces and presentation (e.g., HCI): User Interfaces - Ergonomics

Introduction

Sedentary behavior such as prolonged sitting is applicable to most of us throughout our lives; it applies to K-12 students, college students, and also office workers. Such universal relevance may explain the persistent interest in the effects of sedentary behavior. It is widely believed throughout the media that sitting for an extensive amount of time can shorten the lifespan, mainly due to the influence on cardio metabolic health [6]. One of the ways that may help reduce the health risks is *breaking* sedentary time. According to [14], these breaks include behaviors as simple as shifting positions from sitting to standing or from standing to walking. Despite being generally well-informed of these risks and ways to prevent or reduce them, people are often unaware of how long they actually go on sitting without a break [12].

There has been research in recent years regarding sensor-based chairs that increase awareness of sitting duration and posture, as well as interactive chairs that break up prolonged sitting periods. These types of custom designed chairs, however, are costly and their functions are limited to those specific chairs [15, 17]. The work presented in this paper provides a more affordable and conveniently portable system for monitoring sitting behavior and encouraging breaks in sitting time. The system consists of an augmented BackJoy—a commercially available orthotic device modified to include sensors—and a companion smartphone application. The application specifically tracks and archives sitting behavior and prominently features interactive virtual pets which are designed to interrupt prolonged sitting periods.

Prior to building our prototype, we contacted two medical specialists to see if our proposed artifact would be useful. From these interviews, we learned that our system would be especially useful if it promotes intermittent stretching while sitting and detects crossed legs or left-right imbalance. We also surveyed and interviewed nine possible end users regarding their sitting habits, desire to break sedentary behavior, and experience with virtual pets. Through this process, we were able to discover the desired functions. We also learned that most participants sit for two to four hours maximum without getting up and experience physical discomfort. Most participants felt a moderate to strong need for an application that tracks their sitting posture, and thought the virtual pet element needed to be compelling and well-designed for it to be motivational.

Related Works

There are three parts to introduce as related works regarding sensor-equipped chairs that promote self-awareness, smartphone applications designed to break sedentary behavior, and virtual pets incorporated for promoting health-related activity. For the sensor-equipped chairs, there is the Ergonomic Balance Seat (e-BASE) Chair. This chair is an ergonomic office chair that has an alarm function for unbalanced sitting postures. The alarm is triggered by detecting the uneven pressure distribution of seat pan and backrest [15]. Another is the interactive chair that promotes activity in office environments. This chair rotates, tilts, bounces as an input device of the computer [18]. As for the smartphone application, most are designed to break sedentary behavior by simply alerting the users to stretch with stretching images that many users found dull [5, 13, 19]. One of the more interactive applications uses the embedded accelerometer in the

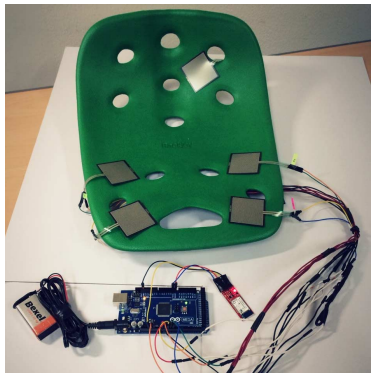
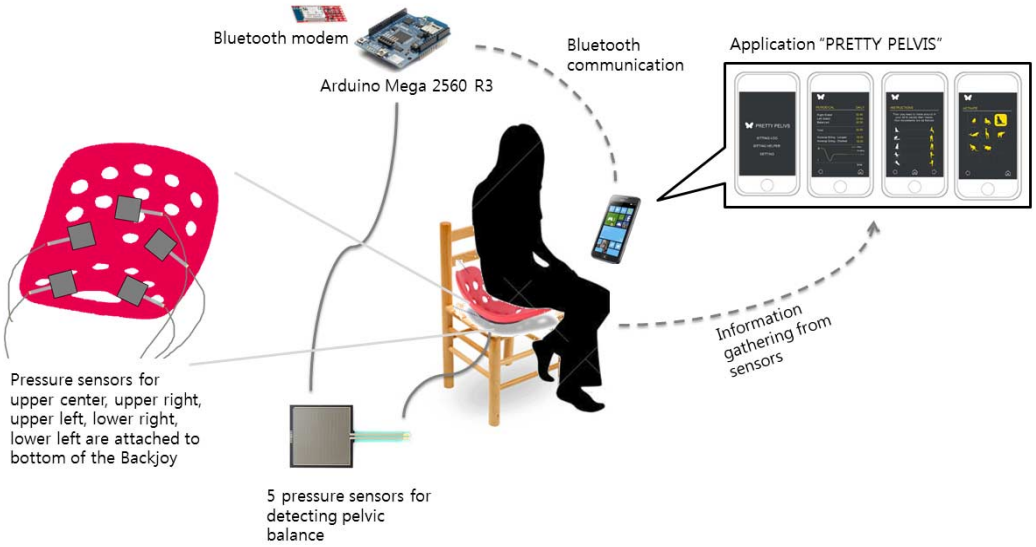


Figure 1. BackJoy with pressure sensors

Figure 2. Interaction map of the hardware components



phone to measure the tilt of the upper body and notifies the user with an alarm [1]. Lastly, one of the earliest attempts to use virtual pets to promote physical activity was the Pocket Pikachu developed by Nintendo in 1998. This digital pet is designed to persuade exercise using an in-built pedometer [7]. With constant interest in promoting physical activity through virtual pet, there have been researches and projects that prevent childhood obesity [8, 9]. Some virtual pet devices target a specific group such as asthma patients [11] or autistic patients [2].

Our system is unique first in terms of the equipment used to make the system. BackJoy, a portable seat device that supports good sitting posture [3], is augmented to collect sitting-related data and to transmit that information to a smartphone. Secondly, our smartphone application has a distinctive motivating factor for breaking sedentary behavior. It incorporates a virtual pet—a personification of a machine with which people can develop an interpersonal relationship [4].

Hardware Components

As a base for building our smart seat device, a small-size BackJoy® SitSmart Posture Plus was chosen for its portability and adaptable form. We also found it advantageous to develop a system that extends an existing reputable product that already enjoys widespread use as an affordable orthotic aid for sitting [16].

To implement a lightweight and robust system that ensures quick and reliable data communication between the seat device and the users' smartphone, we carefully chose our materials. Five square-type pressure sensors are strategically placed and attached onto the seat's surface. The sensor configuration was designed for efficient coverage and detection of postural imbalances, based on consultations with osteopathic and oriental medical doctors. The sensor's shape allows for wider coverage than circular ones, and its size is sufficient for accurate readings without being physically intrusive (see Figure 1). Data from these sensors are sent once every second to the Arduino Mega 2560 R3, a minimal micro-controller-based prototyping board with an internal flash memory (256KB) large enough for robust Bluetooth communication. The Arduino interprets the sensor inputs to extract necessary pertinent information and periodically transmits this data to the smartphone application via Bluetooth (see Figure 2).

Application Development

A prototype smartphone application called *Pretty Pelvis* was developed using Adobe Flash CS6. The application interprets the stream of data from the Arduino to

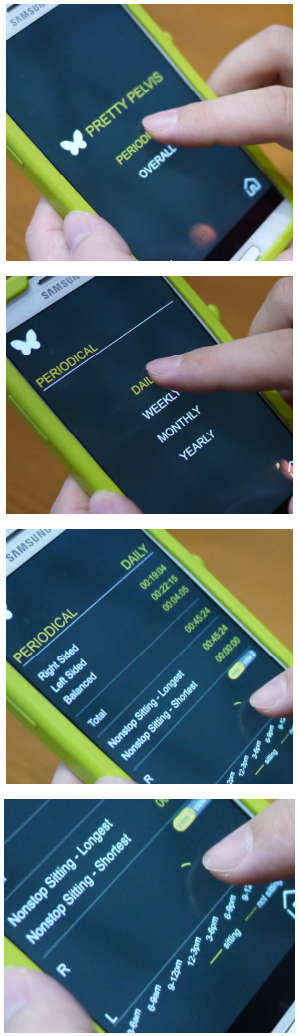


Figure 3. *Sitting Log* of *Pretty Pelvis*

determine the users' pelvic balance, gestural input, and sitting patterns based on programmed parameters. The extracted information is then stored into a database, archived and continually updated.

Users can access their sitting records through the *Sitting Log*, which provides a quick summary and graphic representation of the data by time intervals (daily, weekly, monthly, yearly) in addition to the overall log of all accumulated records. Periodical views consist of an overview and visualization of prolonged sitting periods and breaks as well as visual indications as to whether users were leaning to the left or right (see Figure 3).

The virtual pet function can be activated in the *Sitting Helper* settings, which allow user notifications and a range of alarms to be triggered by prolonged periods of sitting or postural imbalance. The alarms take the visual form of the user's virtual pet animated to express a certain need, which can only be fulfilled when users provide the appropriate physical response for the requisite amount of time. These pet interactions are designed to help break sedentary behavior and correct poor sitting posture in many playful ways.

The Sitting Helper

Our central focus in developing the application was in effectively incorporating virtual pet interaction to act as an incentive for users to break sedentary behavior. Studies have shown that users can form strong personal attachments to their virtual pets with time, due to its humanlike attributes [4]. Since the pet's growth and health is entirely dependent on the frequency of interaction with the owner, virtual pets as toys can cultivate a sense of responsibility and encourage users to "develop and practice nurturing tendencies" [4].The heightened attachment users can develop to their virtual pet supports prolonged engagement [7], which is why virtual pets are increasingly employed in health games as a means to change behavior [17]. Our application's *Sitting Helper* function is its pivotal and defining feature that attempts to utilize the potential of virtual pets as 'persuasive social actors' [10] to promote healthier sitting habits. Our aim was to help users identify with their pets and have all user-pet interactions benefit both parties. When the application detects a prolonged period of postural imbalance or sitting without breaks (users may specify the duration), the user is notified and the pet helper pops up on the screen to incite the user to take action. When the user responds with the appropriate physical action, the pet's state of health is gradually restored. This process is explained more in depth below.

Sitting Helper's status bar

The state of health is represented by five different status levels, which respectively correspond to a specific type of sedentary behavior (see Figure 4).

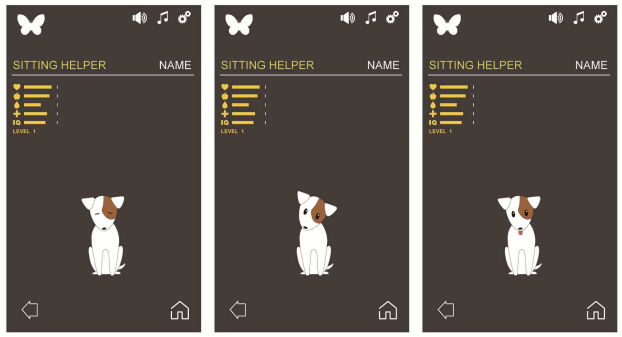


Figure 4. The various idle modes (blinking, head-rotating, panting) and the five status bars

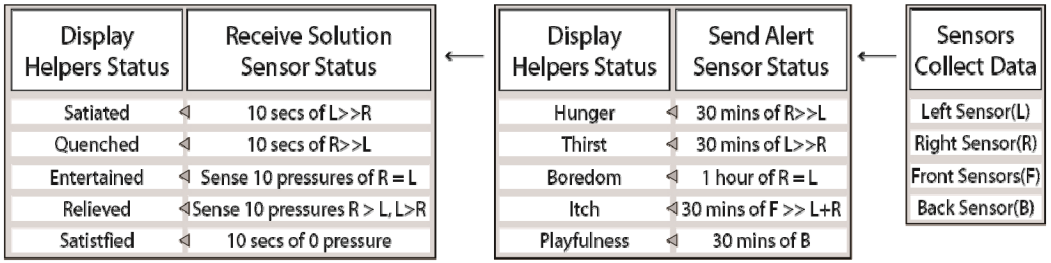


Figure 5. Overall systematic flow of the *Sitting Helper*

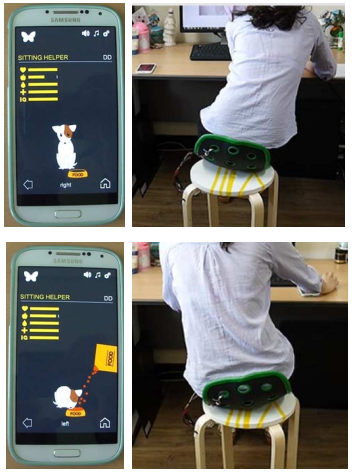


Figure 6. Scene of actual interaction for satisfying the hunger status

The application analyzes the input from the seat's five different sensors to determine whether the user is engaging in any undesirable behavior. The longer the user engages in that behavior, the more the pet's corresponding status is negatively affected. For instance, when sensors on the right show a much higher pressure than ones on the left (indicating the user is leaning to the right), the status bar for hunger decreases. The bar level continues to decrease at a steady rate. The helper stays in idle mode until any of the bars reach a critical low, which triggers the associated alarm.

User-Sitting Helper interactions

After receiving the alarm, users must physically interact with their seat and thus break their sedentary behavior to satisfy their pet's needs. The gestural actions include tilting to the left or the right, bouncing up and down the seat, rocking sideways, and standing up for a set period of time (see Figure 5). These actions are designed to counteract the behavior that caused the warning in the first place. To satisfy the hunger, for example, which was caused by users leaning to the right for too long, users must shift their weight to the left to pour food into their pet's bowl. The pet animation that is shown during the interaction mirrors the gestural action (see Figure 6).

Conclusion & Future Work

In this paper, we explained how the hardware components of our interactive portable chair connects to our smartphone application, and how the virtual pet acts as a persuading social actor for prolonged use and engagement toward the breaking of sedentary behavior. We then gave details on the functions and the systematic flow of the application with emphasis on the *Sitting Helper.*

For future work, we plan on conducting a more comprehensive user test that evaluates the system's usefulness, usability, learnability and satisfaction. Ideally, the test will help statistically prove the following: 1) if the virtual pet function has the intended effect, 2) if *Pretty Pelvis* helps users alleviate back pain, and 3) if users find the application causing reverse effect of becoming more sedentary. We plan to use the evaluation results to update our prototype to offer a better user experience. In the long run, we hope to achieve the status of Internet of Things by connecting the *Pretty Pelvis* system with hospitals that provide orthopedic treatments.

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